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N + - Network Fundamentals & Building Networks.

Weekly Practice and Troubleshooting.

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TASK 1.

❖ Use Cisco Packet Tracer to simulate a WAN setup connecting two branch offices via a VPN tunnel and configure NAT on the edge routers so both branches can access the internet.

- **WAN Setup with VPN Tunnel and NAT:**

IP Address Configuration:

Device	IP Address
Branch 1 LAN	192.168.1.1/24
Branch 2 LAN	192.168.2.1/24
Router 1 WAN	203.0.113.1/30
Router 2 WAN	203.0.113.2/30

Configure VPN Tunnel (GRE):

Router 1:

interface tunnel 0
ip address 10.10.10.1 255.255.255.252
tunnel source g0/1
tunnel destination 203.0.113.2

Router 2:

interface tunnel 0
ip address 10.10.10.2 255.255.255.252
tunnel source g0/1
tunnel destination 203.0.113.1

Configure NAT/PAT

Router 1:

access-list 1 permit 192.168.1.0 0.0.0.255
interface g0/0
ip nat inside
interface g0/1
ip nat outside
ip nat inside source list 1 interface g0/1 overload

Router 2:

access-list 1 permit 192.168.2.0 0.0.0.255
interface g0/0
ip nat inside
interface g0/1
ip nat outside
ip nat inside source list 1 interface g0/1 overload

Verification:

show ip nat translations
show interfaces tunnel 0

Result: Both branch offices are connected through a GRE VPN tunnel, and NAT/PAT allows their internal PCs to access the internet.

TASK 2.

❖ Configure Port Address Translation (PAT) on a router in your simulated WAN so that three devices from the internal network can browse the internet using a single public IP address, Use the 'overload' keyword in your NAT configuration.

- **Configure PAT for Three Internal Devices:**

IP Address Configuration

- PC1: 192.168.1.10/24
- PC2: 192.168.1.20/24
- PC3: 192.168.1.30/24
- Router G0/0: 192.168.1.1/24
- Router G0/1: 203.0.113.1/24

Configure PAT on the Router:

access-list 1 permit 192.168.1.0 0.0.0.255

interface g0/0
ip nat inside
exit

interface g0/1
ip nat outside
exit

ip nat inside source list 1 interface g0/1 overload

Verify PAT:

show ip nat translations

Result: All three internal devices can access the internet using a single public IP address through PAT.

TASK 3.

❖ Intentionally misconfigure the VPN tunnel's pre-shared key between two routers in your WAN simulation, then use the 'show crypto isakmp sa' and 'show run' commands to identify and fix the connectivity issue.

- **Misconfigure the Pre-Shared Key:**

  On Router 1:

  crypto isakmp key wrong123 address 203.0.113.2

  On Router 2:

  crypto isakmp key correct123 address 203.0.113.1

  Identify the Issue:

  show crypto isakmp sa
  show running-config

  The different pre-shared keys cause VPN authentication to fail.

  Fix the Configuration:
  Set the same key on both routers:

  crypto isakmp key correct123 address 203.0.113.2

  On Router 2:

  crypto isakmp key correct123 address 203.0.113.1

Result: The VPN tunnel connectivity is restored after matching the pre-shared keys on both routers.

TASK 4.

❖ Troubleshoot a scenario where devices in one branch office can ping the internet but cannot access devices in the other branch via the VPN tunnel; document the commands you use and the configuration change that resolves the problem.

- **Troubleshoot VPN Connectivity Between Branch Offices:**

Troubleshooting Commands
On both routers, use:

show ip route
show interfaces tunnel 0
show running-config
ping 10.10.10.2

Problem Identified

The static route to the other branch's LAN is missing.

Configuration Fix
On Router 1:

ip route 192.168.2.0 255.255.255.0 10.10.10.2

On Router 2:

ip route 192.168.1.0 255.255.255.0 10.10.10.1

Result: Both branch offices can now communicate through the VPN tunnel after adding the missing static routes.

TASK 5.

❖ Take the WAN, VPN, and NAT scenario you built and swap your configuration file with a classmate; review their setup, find one misconfiguration, and write a short note on how you identified and fixed it, Focus on using 'show' and 'debug' commands for troubleshooting rather than guessing the issue.

- **Troubleshoot WAN, VPN and NAT Configuration:**

  Commands Used:

  show ip route
  show ip nat translations
  show crypto isakmp sa
  show running-config
  debug crypto isakmp
  debug ip nat

  Misconfiguration Found:

  The NAT access list was missing the internal network, so its devices could not access the internet.

  Configuration Fix:

  access-list 1 permit 192.168.1.0 0.0.0.255
  ip nat inside source list 1 interface g0/1 overload

  Result: The NAT configuration was corrected, and the internal devices could access the internet. The VPN connection was also checked using the show and debug commands.